Step 1: Locate and Copy Your Trained Model

You only need one file from your current PC:

- **The weights file:**

D:\RSS Grader\Panisthalam\runs\classify\rubber_run\rss4_vs_rss5-2\weights\best.pt

Copy best.pt to a USB drive, cloud storage (Google Drive), or transfer it over your local network. Rename it to something recognizable like **rubber_grade_best.pt.**

Step 2: Running on Another Laptop / PC (Windows, Mac, or Linux)

1. **Install Python and required libraries on the new machine:** Open the terminal (or Command Prompt) on the new device and run:

Bash

```
pip install ultralytics opencv-python
```

2. **Set up the script:** Create a folder on the new computer, place **rubber_grade_best.pt** inside it, and save this clean deployment script as **run_grading.py**:

Python

```
import cv2
import time
from ultralytics import YOLO

# Load your model from the local folder
model = YOLO("rubber_grade_best.pt")

# If using your phone's IP camera, set its URL:
# (Ensure the new PC and phone are on the same Wi-Fi)
SOURCE = "http://192.168.1.15:8080/video"

# Note: If using the new device's built-in webcam instead, change
to:
```

```
# SOURCE = 0

cap = cv2.VideoCapture(SOURCE)
CONFIDENCE_THRESHOLD = 88.0
prev_time = time.time()

while cap.isOpened():
    ret, frame = cap.read()
    if not ret:
        break

    results = model(frame, imgsz=448, verbose=False)
    probs = results[0].probs
    top_conf = probs.top1conf.item() * 100
    detected_class = results[0].names[probs.top1]

    # FPS
    curr_time = time.time()
    fps = 1.0 / (curr_time - prev_time) if (curr_time - prev_time)
> 0 else 0
    prev_time = curr_time

    if top_conf < CONFIDENCE_THRESHOLD:
        status_text = f"No Sheet Detected ({top_conf:.1f}%)"
        banner_color = (80, 80, 80)
        text_color = (200, 200, 200)
    else:
```

```
        status_text = f"Grade: {detected_class} ({top_conf:.1f}%)"
        banner_color = (0, 100, 0) if detected_class == "RSS4"
else (0, 70, 180)
        text_color = (0, 255, 0) if detected_class == "RSS4" else
(0, 165, 255)


    h, w, _ = frame.shape
    cv2.rectangle(frame, (0, 0), (w, 75), banner_color, -1)
    cv2.putText(frame, status_text, (20, 48),
cv2.FONT_HERSHEY_SIMPLEX, 1.0, text_color, 2)
    cv2.putText(frame, f"FPS: {fps:.1f}", (w - 140, 48),
cv2.FONT_HERSHEY_SIMPLEX, 0.7, (220, 220, 220), 2)


    cv2.imshow("Rubber Sheet Inspector", frame)
    if cv2.waitKey(1) & 0xFF == ord('q'):
        break


cap.release()
cv2.destroyAllWindows()
```

3. **Run it:**

Bash

```
python run_grading.py
```